CLEANASAURUS · ALPHA

"STORAGE ALMOST FULL."

One developer ran a cleanup tool on their laptop and found **58.68 GB** in **128** old `node_modules` folders.

Most of them can be rebuilt with one command. Almost nobody deletes them, because deleting the **wrong** folder the night before a deadline costs you your project.

Source: Flavio Copes, ”npkill” (flaviocopes.com/npkill). Nate Kim · Nate Park · Wesley Chang

MVP VALIDATION · n = 10 (5 CS, 5 non-CS) · planned 15 · Oct 30 – Nov 2

Testers let it act. It didn't get them their space back.

KPI, AS PITCHED IN OUR MVP	PASS	FAIL	RESULT	
Approve and run a **full** plan in session 1	≥ 60%	< 40%	**30%** (3/10) 70% ran at least 80% of a plan	FAIL
Run a second cleanup on their own	≥ 40%	< 20%	**30%** (3/10) measured over 3 days, not 7	ITERATE
Executed actions later undone	≤ 10%	> 25%	**15%** (41 of 267, all from one tester)	ITERATE
Unsafe actions reaching a user	0	any	**0**	PASS

- **Space:** testers wanted 27.5 GB back (median) and got about **2%** of it.
- **A bug in our own number:** FREED said 12.4 GB, but **0 bytes** left the disk. "Deleted" copies sat in a bin on the same drive.
- **Trust slips:** undo was all-or-nothing (C3 dragged files back by hand); C4's own app was flagged as malware; N3 approved 41 actions unread, then undid them all.
- **Limits:** 10 friends and family, not 15; one tester moves a result by 10 points.

PIVOT: CLEAN DEVELOPER CLUTTER FOR CS STUDENTS

Every CS tester pointed at the same place: build folders.

5 of 5

CS testers said, unprompted, that their real space was somewhere the MVP wouldn't go (7 of 10 overall)

38 GB vs 1.4 GB

median per CS tester: rebuildable clutter vs. duplicates the MVP could remove (about 27× more)

WHERE CS TESTERS' 192 GB WAS	GB	HOW IT COMES BACK
Xcode build data, simulators	57	Xcode rebuilds it on the next build
ML model caches	44	re-downloads when the model loads
JavaScript `node_modules`, npm cache	26	`npm ci` re-downloads it
Docker images, build cache	25	`docker pull`
Python environments, caches	22	`pip install -r requirements.txt`
Unity `Library/` folders	18	Unity rebuilds it when the project opens

Rebuildable = one command re-creates it, so deleting it loses nothing, as long as we prove the project can rebuild it. **Why CS students:** they met all 3 trust KPIs, while non-CS testers undid 37% of actions and 3 of 5 called it too technical. Tools like CleanMyMac's CLI delete by age; we delete only what we can prove comes back.

DEMO · VIDEO, NARRATED LIVE · 3½ MIN

▶ PLAY THE DEMO

1. An end-of-semester `Downloads` folder: lecture PDFs, duplicate downloads, files you don't remember saving
2. One sentence: **organize my school PDFs by class**, **remove exact duplicates**, **quarantine anything the scanner flagged**, and **leave my résumés alone**
3. Review the plan: each move shows where and why. Duplicate `index.js` files are left alone (a project may need each copy). We uncheck one PDF and approve the rest
4. The PDF lands in its class folder, then **UNDO** reverses the whole batch

WHAT YOU JUST SAW

I describe it. Claude proposes. The app checks. I decide.

Claude proposed

- Which class each PDF belongs to, from clues in its filename
- Which copies to remove and which flagged files to quarantine
- The exception we asked for: résumés stay where they are
- A plain-English reason for every action, including why duplicate `index.js` files should stay

The app checked

- A duplicate is removed only if an identical copy survives the cleanup
- A file is quarantined only if the scanner actually flagged it
- A failed check goes back to Claude to revise the plan
- Nothing runs until we approve; we can uncheck any line, and UNDO reverses the batch

METHODOLOGY & ENGINEERING METRICS · measured Nov 5 on main (787454f)

29

user stories
each one is a GitHub issue (#2–#30), with acceptance criteria

29 / 18

issues logged / closed
6 built in the MVP + 12 in the Alpha; 11 open, 1 of them partial · `gh issue list`

129

automated tests
`npm test` · plus 15 click-through UI steps and 5 live tests on the real Claude (5/5)

4.4

tests per story
129 tests ÷ 29 stories · every story has at least 2

100%

of runnable tests pass
101 of 101 · 78% of all 129: the other 28 are kept red for stories not built yet

95.9%

backend line coverage
`npm run test:coverage` · 90.2% of branches

25 : 0

AI-written tests changed after review : tests hand-edited by a teammate so far
every change is logged in tdd/README §5 · team review scheduled before Beta

Tests first, no code (c1c337d) → Rebuild proof green (d2f885d, 454d384) → Claude-path + intent tests red → 101/101 green → Live Claude 5/5

93 of 129 tests (72%) were written before the code they test. Review produced 19 findings: 7 real bugs, 5 missing tests, 5 other test defects, and 2 tests that couldn't fail. CI fails if a test for an unbuilt story ever passes by accident.

STAGE & ROADMAP TO BETA

Alpha: done

✓ Validation run (n = 10), decision: Pivot
✓ 29 stories → 29 GitHub issues with acceptance criteria
✓ Rebuild proof + JavaScript and Python cleanup, real deletes
✓ Honest FREED, measured on disk
✓ Clean by intent, caches, conda environments
✓ One-click rebuild, undo one action, macOS Trash, one-line summary
✓ Claude path: 7 simulated + 5 live tests (5/5); CI on every push

Beta: four goals, each with a pass line

GOAL	PASS IF
1. Real space, simpler removal. The Dino Bin goes away; removing files frees space you can see.	≥ 70% of testers see storage actually freed on disk
2. One question, not twenty. Instead of asking about each file, Claude asks one question that settles all similar copies and deletable files.	≥ 60% say deleting felt smoother than in the Alpha and the questions were satisfying
3. Find what people actually want gone. A narrower scope plus built-in hints, e.g. CS students want `node_modules` from inactive JS projects deleted.	≥ 80% say what they wanted cleaned was cleaned, and run ≥ 80% of the plan
4. Built for CS students. Claude starts knowing what to look for (Python `.venv`, pip caches), hard rules (never source code, git repos, uncommitted work, `.env` files), developer language, and a read-only summary of the dev setup.	≥ 60% of CS testers say it fits their needs better

QUESTIONS?

Claude writes the plan. Plain code proves it’s safe. You get the space back.

github.com/natekim416/cs4501_startup · tests: `cd app && npm test`